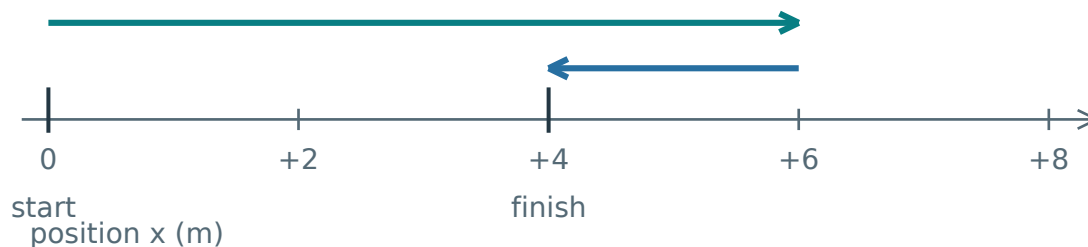


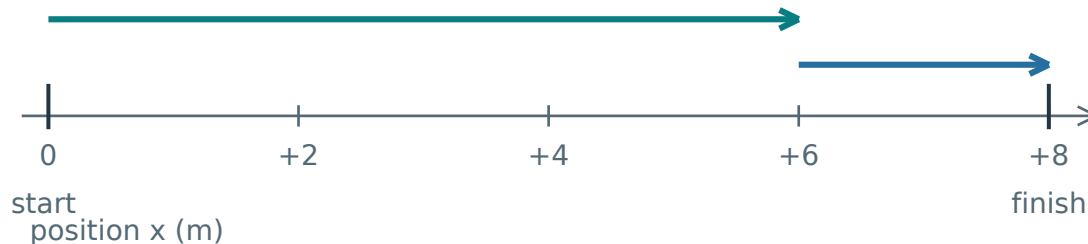
The same path length can end in different places

Both journeys start at 0 m and cover 8 m. Compare what changes.

A: 6 m right, then 2 m left



B: 6 m right, then 2 m right



In A, part of the route is retraced. In B, every part contributes to the same rightward change.

Compare the two totals

Both distances: $6 + 2 = 8$ m.

Displacement A: $+6 - 2 = +4$ m.

Displacement B: $+6 + 2 = +8$ m.

Reconstruct the rule

$$\Delta x = x_f - x_i$$

Δx means change in position; x_i is the initial position and x_f the final position.

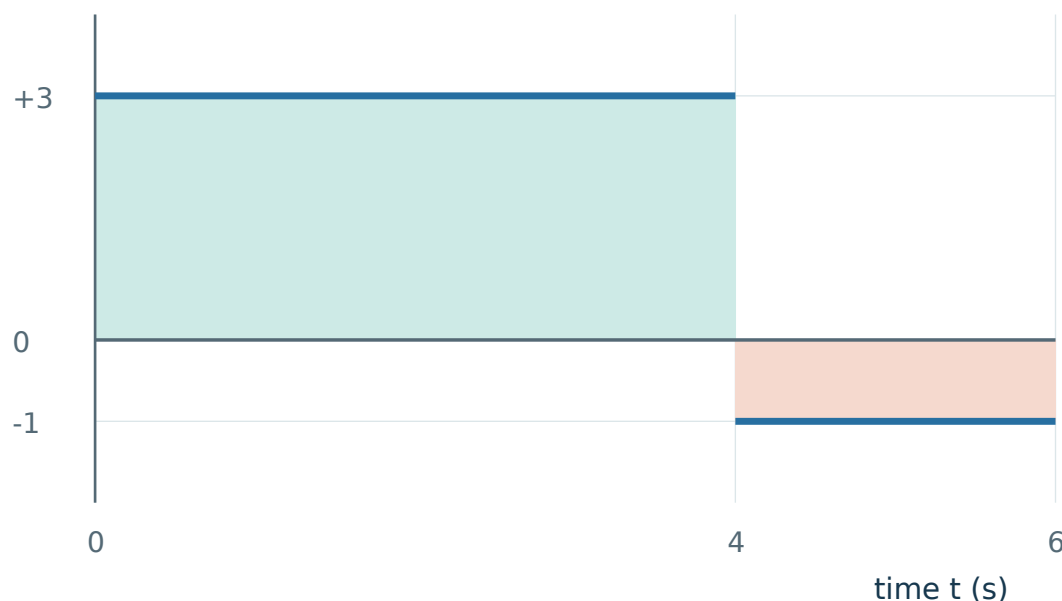
Distance adds path lengths. The bars in $|\Delta x|$ mean size without the sign. This size cannot exceed the path length; equality holds with no reversal.

Changing the origin changes both coordinates equally, so their difference is unchanged. Reversing the positive axis changes the sign of Δx , but not the distance.

Read a graph as accumulated movement

Take right as positive. A constant velocity gives one rectangle of movement.

velocity v (m/s)



Areas are +12 m and -2 m. The second width is $6 - 4 = 2$ s, not 6 s.

Build from a constant interval

$$\Delta x = v \Delta t$$

Velocity is signed movement per second; multiplying by elapsed time reconstructs the movement.

Thus graph height $\times$ width gives displacement. Units: $(\text{m/s}) \times \text{s} = \text{m}$.

The abrupt switch at $t = 4$ s is an idealisation. It contributes no extra time or distance.

$$\Delta x = +12 - 2 = +10 \text{ m}$$

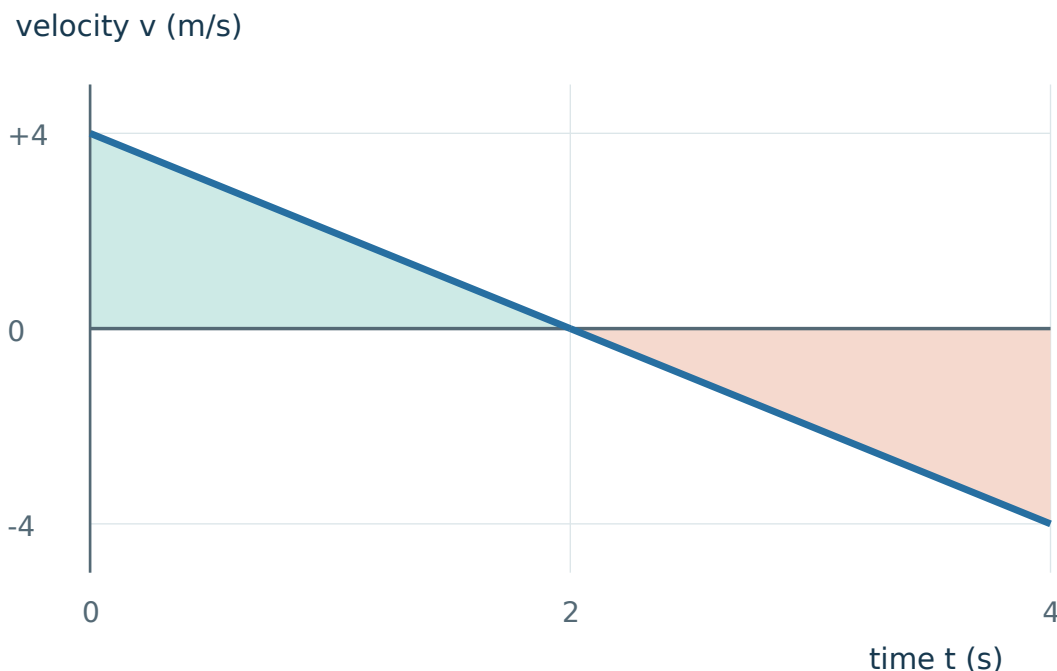
$$\text{Distance} = 12 + 2 = 14 \text{ m}$$

Final position still needs a start: $x_f = x_i + \Delta x$. A v - t graph alone does not provide x_i .

Keep graph height, slope and area separate: height is velocity; slope is its rate of change; signed area is displacement.

A straight segment can cross zero

Velocity falls linearly from +4 m/s at 0 s to −4 m/s at 4 s.



Each triangle has base 2 s and height size 4 m/s. The sign changes at 2 s.

Predict before calculating

The two equal area sizes cancel in displacement. Both still count as movement.

$$\Delta x = +\frac{1}{2}(2)(4) - \frac{1}{2}(2)(4)$$

$$= 0 \text{ m}$$

$$\text{Distance} = 4 + 4 = 8 \text{ m}$$

The $\frac{1}{2}$ comes from triangular geometry. Straight graph segments let us calculate these areas exactly.

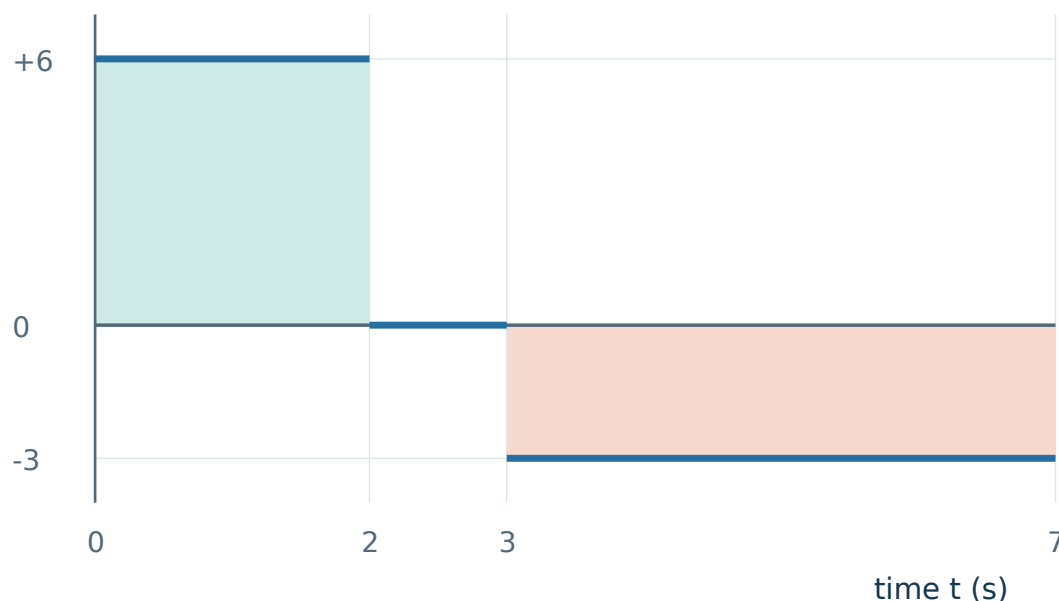
Zero displacement can hide substantial movement. Returning to the start does not imply continuous rest.

A zero value at an instant is not sufficient to prove a reversal. Check whether velocity actually changes sign on the two sides of that instant.

A stopped interval changes time, not path length

A sensor records +6 m/s for 2 s, rest for 1 s, then −3 m/s for 4 s.

velocity v (m/s)



Treat each interval honestly

The changes are +12 m, 0 m and −12 m. Rest contributes zero area, even though time passes.

$$\Delta x = +12 + 0 - 12 = 0 \text{ m}$$

$$\text{Distance} = 12 + 0 + 12 = 24 \text{ m}$$

Change the starting position

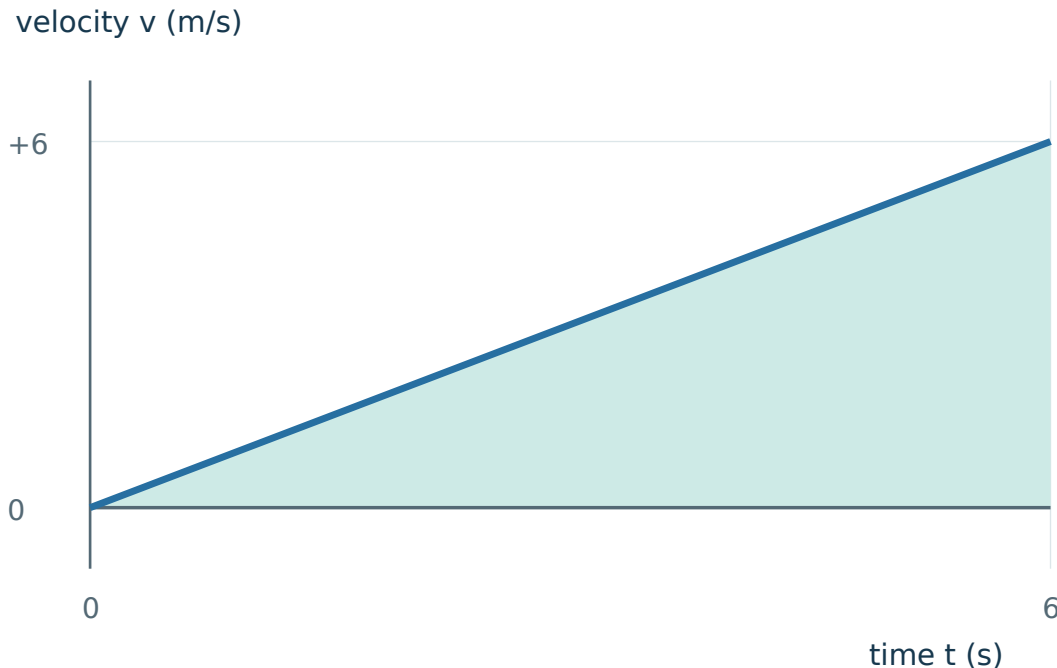
Start at +9 m and finish at +9 m. The same areas still give zero displacement and 24 m distance.

Movement: 12 m right, no movement for 1 s, then 12 m left. The return ends at the start.

Check a solution using both representations: a signed-area calculation and a right-stop-left motion sketch must tell the same story.

Why the final height can mislead you

Velocity rises linearly from 0 to +6 m/s over 6 s.



The actual area is half the rectangle formed by the final velocity and the full duration.

Test the tempting shortcut

Using 6×6 assumes the object travels at +6 m/s throughout. That contradicts the graph.

$$\Delta x = \frac{1}{2} \times 6 \times 6 = +18 \text{ m}$$

Distance is 18 m because there is no negative-velocity interval.

What if the line is curved?

Do not force it into a triangle. Signed area still represents displacement, but its numerical evaluation needs more information or an approximation.

For independent practice, choose your own diagram. A shorter explanation is acceptable only if the model, signs and physical check remain clear.

Appendix A · Questions

Use the diagrams you need. A correct number needs a physical explanation.

A1 · Apply

Start at +1 m. Walk 5 m left, then 3 m right. Find the final position, distance and displacement.

Start → turning point → finish

Final position: ____

Distance: ____ Displacement: ____

Hints · p. 11

Solution · p. 13

A2 · Apply

An object moves from -2 m to $+5$ m without reversing. Find its distance and displacement. Explain the signs.

Sketch the line and explain why crossing zero is not a turn.

Hints · p. 11

Solution · p. 13

Appendix A · Questions

Use the diagrams you need. A correct number needs a physical explanation.

A3 · Explain

Two students start at 0 m and finish at +4 m. Must they have travelled the same distance? Draw two journeys to support your answer.

Draw and label two routes. Keep the endpoints the same.

Hints · p. 11

Solution · p. 14

A4 · Transfer

A particle goes from -3 m to $+5$ m, then returns to $+1$ m. Find distance and displacement. A second observer adds 10 m to every coordinate. What changes?

Show the route or working that makes your answer clear.

Hints · p. 11

Solution · p. 14

Appendix A · Questions

Use the diagrams you need. A correct number needs a physical explanation.

A5 · Compare

Journey A: $+3 \text{ m/s}$ for 4 s , then $+2 \text{ m/s}$ for 2 s . Journey B changes only the last velocity to -2 m/s . Find both distances and displacements. Explain what stays the same.

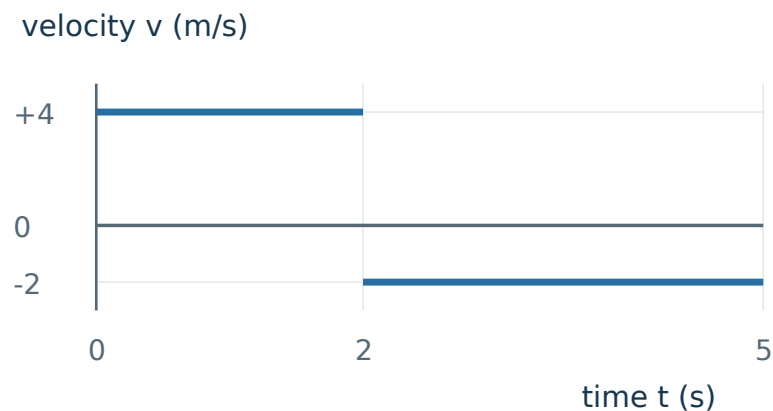
Show the route or working that makes your answer clear.

Hints · p. 12

Solution · p. 15

A6 · Apply

Use the graph to find displacement and distance from 0 s to 5 s . Give a physical meaning for each area.



Hints · p. 12

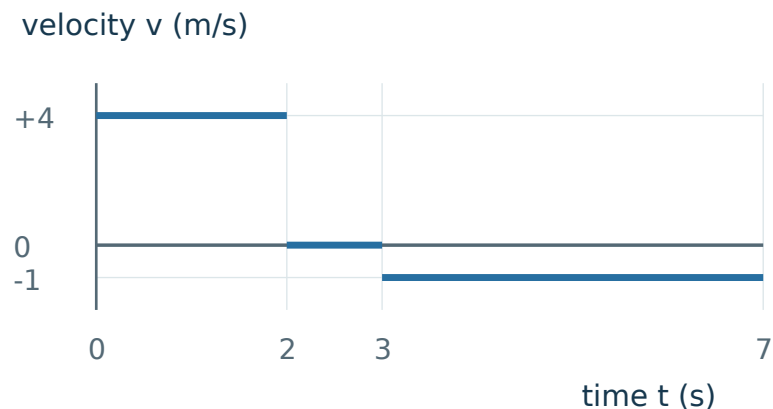
Solution · p. 15

Appendix A · Questions

Use the diagrams you need. A correct number needs a physical explanation.

A7 · Transfer

A robot has the graph shown and starts at $x = -5$ m. Find its distance, displacement and final position after 7 s.

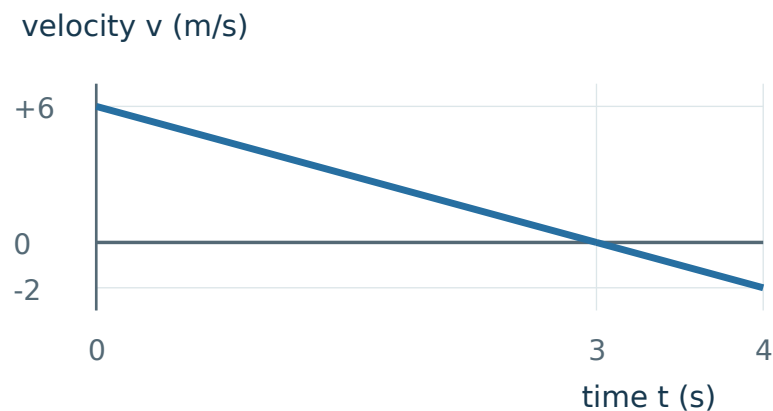


Hints · p. 12

Solution · p. 16

A8 · Transfer

The velocity follows the straight graph shown. Find distance and displacement over 4 s. At what time does the particle reverse direction? Justify this from the graph.



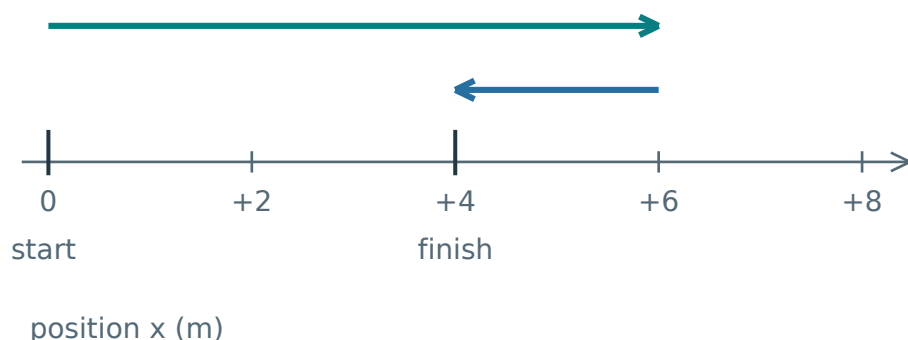
Hints · p. 12

Solution · p. 16

Appendix B · Keep the picture, rebuild the rule

Right is positive. Cover the words and explain each diagram aloud.

1 Distance and displacement



Distance: add every path length.

Displacement: the signed endpoint change.

$$\Delta x = x_f - x_i$$

Δx : displacement; x_i : initial position;

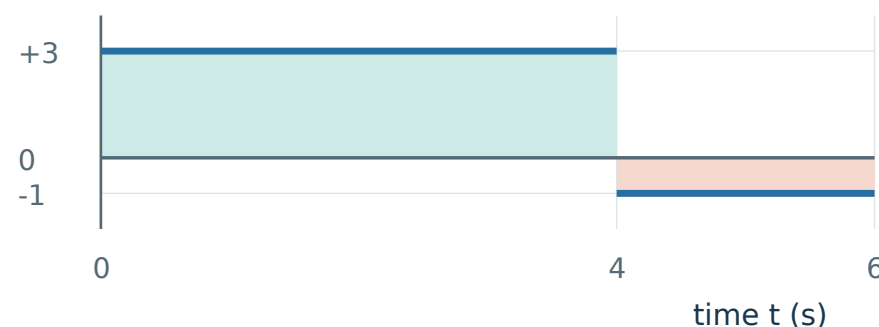
x_f : final position. Here, distance is 8 m and Δx is +4 m.

Check: distance $\geq |\Delta x|$. Bars mean size.

Rebuild: trace the path, then compare its endpoints.

2 Reading a velocity-time graph

velocity v (m/s)



Displacement: add signed areas.

Distance: add all the area sizes. Split at zero crossings.

Rectangle: $v \times \text{duration}$

Triangle size: $\frac{1}{2} \times \text{base} \times \text{height}$.

Graph above: $\Delta x = +10$ m; distance 14 m.

Check: (m/s) $\times$ s = m. Use a rectangle for constant velocity. Use $\frac{1}{2} \times \text{base} \times \text{height}$ only for a triangular region; other straight segments may need rectangle + triangle.

Hints · Read one step, then try again

Cover the rows below the hint you are reading.

- A1**
- H1 Notice** Keep the starting coordinate separate from the lengths walked.
- H2 Model** Mark the start, the turning point and the finish on a number line.
- H3 Start** The turning point is at $+1 - 5$ m. Find the finish next.

[Return · p. 6](#)

- A2**
- H1 Notice** Crossing the origin does not mean reversing direction.
- H2 Model** Draw -2 , 0 and $+5$ in order along the line.
- H3 Start** The first part of the path, from -2 m to 0 m, is 2 m.

[Return · p. 6](#)

- A3**
- H1 Notice** The question fixes the endpoints, not the route.
- H2 Model** Draw one direct route and one route with a turn.
- H3 Start** For the turning route, choose a turning point beyond $+4$ m.

[Return · p. 7](#)

- A4**
- H1 Notice** The second observer changes the labels, not the physical journey.
- H2 Model** Draw the three positions, then label each with 10 m added.
- H3 Start** Compare $(1 + 10) - (-3 + 10)$ with $1 - (-3)$.

[Return · p. 7](#)

Hints · Read one step, then try again

Cover the rows below the hint you are reading.

- A5**
- H1 Notice** Only the direction of the last interval changes.
- H2 Model** Sketch the two graphs with the same scales.
- H3 Start** Calculate the common first area: $(+3 \text{ m/s}) \times (4 \text{ s})$.

[Return · p. 8](#)

- A6**
- H1 Notice** Read the duration of each horizontal segment.
- H2 Model** Separate the two rectangular regions at $t = 2 \text{ s}$.
- H3 Start** For the first region, write $(+4) \times (2 - 0)$.

[Return · p. 8](#)

- A7**
- H1 Notice** The starting coordinate is separate from the displacement.
- H2 Model** Pair the graph regions with right, stopped and left motion arrows.
- H3 Start** The final region has duration $7 - 3 \text{ s}$.

[Return · p. 9](#)

- A8**
- H1 Notice** Locate the zero crossing before calculating areas.
- H2 Model** Use the straight-line change to split the graph into two triangles.
- H3 Start** The velocity decreases by $8 \div 4 \text{ m/s}$ each second.

[Return · p. 9](#)

Solutions · Compare the reasoning, not just the number

If a step surprised you, revisit the linked lesson and explain its picture.

A1 Start +1 m; 5 m left, then 3 m right.

Why. Position changes carry direction; path lengths do not.

Method. Final position = $1 - 5 + 3 = -1$ m. Distance = $5 + 3 = 8$ m. Displacement = $-1 - (+1)$.

Answer / check. Final position -1 m; distance 8 m; displacement -2 m (left).

Keep. The final coordinate is not automatically the displacement.

[Return to A1 · p. 6](#)

[Lesson · p. 1](#)

A2 Move from -2 m to $+5$ m with no reversal.

Why. The route goes 2 m to zero and another 5 m right.

Method. Distance = $2 + 5$. Displacement = $+5 - (-2)$.

Answer / check. Distance 7 m; displacement $+7$ m (right).

Keep. Equal distance and displacement size require no reversal in this straight-line journey.

[Return to A2 · p. 6](#)

[Lesson · p. 1](#)

Solutions · Compare the reasoning, not just the number

If a step surprised you, revisit the linked lesson and explain its picture.

A3 Same start 0 m and finish +4 m; compare possible distances.

Why. Endpoints fix displacement but leave the path undecided.

Method. One can go directly $0 \rightarrow +4$. Another can go $0 \rightarrow +7 \rightarrow +4$. Their paths are 4 m and $7 + 3$ m.

Answer / check. No. These distances are 4 m and 10 m; both displacements are +4 m.

Keep. A drawing can disprove an “always” claim.

[Return to A3 · p. 7](#)

[Lesson · p. 1](#)

A4 Route $-3 \rightarrow +5 \rightarrow +1$ m; then shift every coordinate by +10 m.

Why. Displacement is a coordinate difference, so a common shift cancels.

Method. Path lengths: $5 - (-3) = 8$ m and $5 - 1 = 4$ m. $\Delta x = 1 - (-3) = +4$ m. Shifted endpoints are +7 m and +11 m.

Answer / check. Distance 12 m and displacement +4 m for both origins. Positions change.

Keep. A reference origin changes position values, not a fixed journey.

[Return to A4 · p. 7](#)

[Lesson · p. 1](#)

Solutions · Compare the reasoning, not just the number

If a step surprised you, revisit the linked lesson and explain its picture.

A5 A: +3 for 4 s, then +2 for 2 s. B: last velocity becomes −2 m/s.

Why. Reversing the second velocity changes the area sign, not its size.

Method. First area is +12 m in both. The last area is +4 m in A and −4 m in B. Add signs for displacement and sizes for distance.

Answer / check. A: $\Delta x = +16$ m, distance 16 m. B: $\Delta x = +8$ m, distance 16 m.

Keep. A sign change can alter the endpoint without altering total path.

[Return to A5 · p. 8](#)

[Lesson · p. 2](#)

A6 Graph: +4 m/s from 0-2 s, then −2 m/s from 2-5 s.

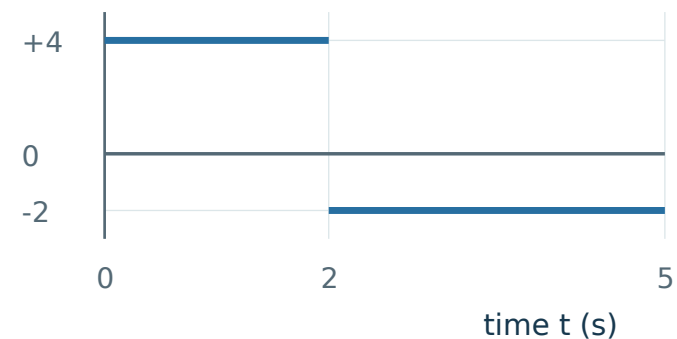
Why. Graph widths are durations. The second width is $5 - 2 = 3$ s.

Method. Signed areas are $+4 \times 2 = +8$ m and $-2 \times 3 = -6$ m. For distance count both path lengths.

Answer / check. Displacement +2 m; distance 14 m. The parts mean 8 m right and 6 m left.

Keep. Read each width from its endpoints before multiplying.

velocity v (m/s)



[Return to A6 · p. 8](#)

[Lesson · p. 2](#)

Solutions · Compare the reasoning, not just the number

If a step surprised you, revisit the linked lesson and explain its picture.

A7 Graph +4 m/s for 2 s, rest 1 s, then -1 m/s for 4 s; start -5 m.

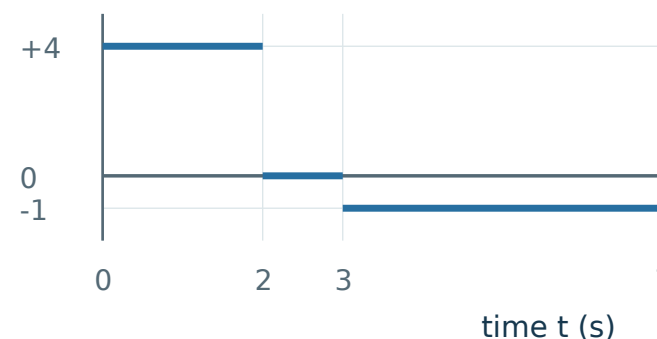
Why. Rest adds no displacement. Final position is start plus signed change.

Method. Areas are +8 m, 0 m and -4 m. Add their sizes for distance and their signed values for displacement. Then add the change to -5 m.

Answer / check. Distance 12 m; displacement +4 m; final position -1 m.

Keep. A positive displacement can finish at a negative coordinate.

velocity v (m/s)



[Return to A7 · p. 9](#)

[Lesson · p. 4](#)

A8 Velocity falls linearly from +6 to -2 m/s over 4 s.

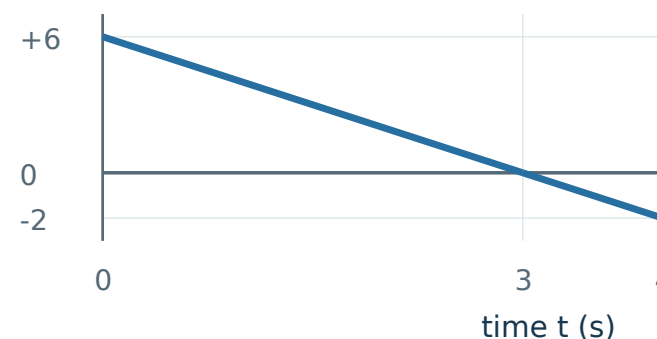
Why. Reversal occurs where velocity changes sign; split the signed area there.

Method. Velocity falls 8 m/s in 4 s. It reaches zero after $6 \div 2 = 3$ s. Areas: $+\frac{1}{2} \times 3 \times 6 = +9$ m; $-\frac{1}{2} \times 1 \times 2 = -1$ m.

Answer / check. Displacement +8 m; distance 10 m; reversal at 3 s.

Keep. Equal endpoint signs are not the test; inspect the sign throughout the interval.

velocity v (m/s)



[Return to A8 · p. 9](#)

[Lesson · p. 3](#)